THE TWISTED INVERSE IMAGE OF AN ETALE MORPHISM AGREES WITH PULLBACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Section on finite support lower shriek (Section section-finite-support, canonical tag 0F6E). cited at proof lines 2460-2461. Where the j_! whose right adjoint is j^! is defined in this chapter.
- Extension by zero section (Etale Cohomology, Section etale-cohomology-section-extension-by-zero, canonical tag 03S2). cited at proof lines 2462-2463 and 2465-2466. The other definition of j_!, as the left adjoint of j^{-1}.
- Comparison of the two j_! for etale morphisms (Lemma lemma-finite-support-etale-shriek, canonical tag 0F6Q). cited at proof lines 2463-2464. Identifies the two definitions of j_!.

Source reference. Original section 0F6E: sources/stacks-chow/tags.txt, source lines 18969–18969. Cited at proof lines 2460-2461: where the j_! whose right adjoint is j^! is defined in this chapter.

Source reference. Original section 03S2: sources/stacks-chow/tags.txt, source lines 4951–4951. Cited at proof lines 2462-2463 and 2465-2466: the other definition of j_!, as the left adjoint of j^{-1}.

Source reference. Original lemma 0F6Q: sources/stacks-chow/tags.txt, source lines 18980–18980. Cited at proof lines 2463-2464: identifies the two definitions of j_!.

Lemma 0.2. *Let $j : U \to X$ be an étale morphism. Then $j^! = j^{-1}$.*

Proof. This is true because $j_!$ as defined in Section 0F6E agrees with $j_!$ as defined in Étale Cohomology, Section 03S2, see Lemma 0F6Q. Finally, in Étale Cohomology, Section 03S2 the functor $j_!$ is defined as the left adjoint of j^{-1} and hence we conclude by uniqueness of adjoint functors. □

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